

Bateria PF AC — Comparação (14)–(19)

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1. Tabela Resumo Global de Convergência e Clusters

Status abreviados: **OK**=LOCALLY_SOLVED · **INF**=INFEASIBLE · **ITL**=ITERATION_LIMIT · **TIM**=TIME_LIMIT · **KIL**=TIMEOUT_KILLED · **CRA**=SUBPROCESS_CRASH · **ERR**=OTHER_ERROR

Caso	(14)	(15)	(16)	(17)	(18)	(19)	Clusters	Observações
01 MAXIMA NOTURNA_DEZ25.PWF	KIL	ERR	KIL	ERR	KIL	ERR	—	Nenhuma formu- lação convergiu
300bus.pwf	ERR	OK	OK	OK	OK	OK	1,2,3,4,5	1/6 não convergi- ram
3bus.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
3bus_DBSH.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
3bus_DCER.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
3bus_DCSC.pwf	OK	OK	OK	OK	OK	OK	1,2,3,4	
3bus_DCline.pwf	ERR	ERR	OK	OK	OK	OK	1,2	1/6 não convergi- ram
3bus_DSHL.pwf	ERR	OK	OK	OK	OK	OK	1,2,3	1/6 não convergi- ram
3bus_correction.pwf	ERR	ERR	ERR	ERR	ERR	ERR	—	Nenhuma formu- lação convergiu
3bus_shunt_fields.pwf	OK	OK	OK	OK	OK	OK	1,2,3,4	
3busfrank.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
3busfrank_continuous.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
3busfrank_q.pwf	OK	OK	OK	OK	OK	OK	1,2,3,4	
4busfrank_v.pwf	ERR	ERR	OK	OK	OK	OK	1,2,3	2/6 não convergi- ram
500bus.pwf	ERR	OK	OK	OK	OK	OK	1,2,3,4	1/6 não convergi- ram
5busfrank.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
5busfrank_caps.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
5busfrank_c.pwf	ERR	ERR	OK	OK	OK	OK	1,2	2/6 não convergi- ram
5busfrank_cdf.pwf	OK	OK	OK	OK	OK	OK	1,2,3,4	
5busfrank_cdf.pwf	OK	OK	OK	OK	OK	OK	1,2,3,4	
9bus.pwf	OK	OK	OK	OK	OK	OK	1,2,3	
9bus_transformer_fields.pwf	OK	OK	OK	OK	OK	OK	1,2,3,4	
test_default.pwf	ERR	ERR	ERR	ERR	ERR	ERR	—	Nenhuma formu- lação convergiu

Caso	(14)	(15)	(16)	(17)	(18)	(19)	Clusters	Observações
test_line_smbus.pwf	ERR	ERR	ERR	ERR	ERR	ERR	—	Nenhuma formu- lação convergiu
test_system.pwf	ERR	ERR	OK	OK	OK	OK	1,2	2/6 não convergi- ram

2. Análise Barra-a-Barra

Critério: para cada barra, listada apenas se $|\Delta vm|$ ou $|\Delta va|$ em (16)/(17)/(18)/(19) excede 0.0001 em relação a AMBAS (14) E (15). Barras concordando com pelo menos uma das referências PowerModels são omitidas.

Caso 01 MAXIMA NOTURNA_DEZ25.PWF

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 300bus.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(16)	4.6854e-02	4.0219e-02	1.0704	9.2550e-01
1	(17)	4.9243e-02	4.7770e-02	1.0728	9.1794e-01
1	(18)	4.2380e-02	1.0618e-01	1.0659	8.5953e-01
1	(19)	4.2379e-02	1.0749e-01	1.0659	8.5822e-01
2	(16)	4.2901e-02	4.0168e-02	1.0752	9.4032e-01
2	(17)	4.5016e-02	4.7599e-02	1.0773	9.3289e-01
2	(18)	2.7948e-02	1.0164e-01	1.0602	8.7885e-01
2	(19)	2.7948e-02	1.0289e-01	1.0602	8.7760e-01
3	(16)	3.8124e-02	4.1025e-02	1.0215	9.3901e-01
3	(17)	4.3042e-02	4.8802e-02	1.0264	9.3123e-01
3	(18)	2.7166e-02	1.0631e-01	1.0105	8.7372e-01
3	(19)	2.7859e-02	1.0764e-01	1.0112	8.7240e-01
4	(16)	3.9889e-02	3.6027e-02	1.0472	8.8319e-01
4	(17)	4.4989e-02	4.3204e-02	1.0523	8.7601e-01
4	(18)	9.2696e-02	9.6776e-02	1.1000	8.2244e-01
4	(19)	9.2696e-02	9.7990e-02	1.1000	8.2123e-01
5	(16)	4.6792e-02	3.8252e-02	1.0606	9.0497e-01
5	(17)	4.9590e-02	4.5748e-02	1.0634	8.9747e-01
5	(18)	4.1097e-02	1.0429e-01	1.0549	8.3893e-01
5	(19)	4.1029e-02	1.0559e-01	1.0548	8.3763e-01

(mostrando 20 de 1200 barras divergentes; tabela truncada)

Caso 3bus.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(17)	9.2947e-02	1.5046e-35	1.0071	0.0000e+00
1	(18)	9.2947e-02	1.5046e-35	1.0071	0.0000e+00
1	(19)	9.2947e-02	1.5046e-35	1.0071	0.0000e+00
2	(17)	9.2652e-02	6.6894e-03	1.0073	-6.6894e-03
2	(18)	9.2652e-02	6.6894e-03	1.0073	-6.6894e-03
2	(19)	9.2651e-02	6.6894e-03	1.0073	-6.6894e-03

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
3	(17)	9.5820e-02	7.0124e-03	0.9740	-2.5383e-02
3	(18)	9.5820e-02	7.0124e-03	0.9740	-2.5383e-02
3	(19)	9.5819e-02	7.0124e-03	0.9740	-2.5383e-02

Caso 3bus_DBSH.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(17)	1.3044e-01	6.4160e-38	1.0997	0.0000e+00
1	(18)	1.3044e-01	6.4160e-38	1.0997	0.0000e+00
1	(19)	1.3044e-01	6.4160e-38	1.0997	0.0000e+00
2	(17)	1.3044e-01	7.6171e-03	1.0997	-7.6171e-03
2	(18)	1.3044e-01	7.6171e-03	1.0997	-7.6171e-03
2	(19)	1.3044e-01	7.6143e-03	1.0997	-7.6143e-03
3	(17)	1.3000e-01	3.9519e-03	1.0300	1.3387e-02
3	(18)	1.3000e-01	3.9519e-03	1.0300	1.3387e-02
3	(19)	1.3000e-01	3.9532e-03	1.0300	1.3388e-02

Caso 3bus_DCER.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(17)	5.2795e-02	6.4160e-38	1.0221	0.0000e+00
1	(18)	5.2795e-02	6.4160e-38	1.0221	0.0000e+00
1	(19)	5.2795e-02	6.4160e-38	1.0221	0.0000e+00
2	(17)	6.0713e-02	1.7178e-02	1.0300	-1.7178e-02
2	(18)	6.0713e-02	1.7178e-02	1.0300	-1.7178e-02
2	(19)	6.0713e-02	1.7178e-02	1.0300	-1.7178e-02
3	(17)	5.6680e-02	5.7136e-03	0.9567	5.4726e-03
3	(18)	5.6680e-02	5.7136e-03	0.9567	5.4726e-03
3	(19)	5.6680e-02	5.7137e-03	0.9567	5.4725e-03

Caso 3bus_DCSC.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(16)	6.4632e-02	4.4373e-38	1.0352	0.0000e+00
1	(17)	9.3469e-02	4.4373e-38	1.0064	0.0000e+00
1	(18)	9.3469e-02	4.4373e-38	1.0064	0.0000e+00
1	(19)	9.3468e-02	4.4373e-38	1.0064	0.0000e+00
2	(16)	6.3632e-02	2.3972e-04	1.0362	-2.3972e-04
2	(17)	9.3600e-02	2.6434e-04	1.0062	-2.6434e-04
2	(18)	9.3600e-02	2.6434e-04	1.0062	-2.6434e-04
2	(19)	9.3600e-02	2.6434e-04	1.0062	-2.6434e-04
3	(16)	7.9726e-02	1.8699e-02	1.0178	-1.2142e-02
3	(17)	1.0977e-01	1.8054e-02	0.9878	-1.2786e-02
3	(18)	1.0977e-01	1.8054e-02	0.9878	-1.2786e-02
3	(19)	1.0977e-01	1.8054e-02	0.9878	-1.2786e-02

Caso 3bus_DCline.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
2	(16)	2.5494e-06	1.7794e+00	1.1000	1.7794e+00
2	(17)	2.5494e-06	1.7794e+00	1.1000	1.7794e+00
2	(18)	2.5494e-06	1.7794e+00	1.1000	1.7794e+00
2	(19)	2.5494e-06	1.7794e+00	1.1000	1.7794e+00
3	(16)	1.6977e-01	2.0591e-01	0.9000	1.8754e-01
3	(17)	1.6977e-01	2.0591e-01	0.9000	1.8754e-01
3	(18)	1.6977e-01	2.0591e-01	0.9000	1.8754e-01
3	(19)	1.6977e-01	2.0591e-01	0.9000	1.8754e-01

Caso 3bus_DSHL.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(16)	1.6820e-02	0.0000e+00	1.0458	0.0000e+00
1	(17)	1.6842e-02	0.0000e+00	1.0458	0.0000e+00
1	(18)	1.6842e-02	0.0000e+00	1.0458	0.0000e+00
1	(19)	1.6842e-02	0.0000e+00	1.0458	0.0000e+00
2	(16)	2.3782e-01	1.9828e-01	0.9000	1.3356e-01
2	(17)	2.3782e-01	1.9826e-01	0.9000	1.3358e-01
2	(18)	2.3782e-01	1.9826e-01	0.9000	1.3358e-01
2	(19)	2.3782e-01	1.9826e-01	0.9000	1.3358e-01
3	(16)	2.2793e-01	1.5416e-01	0.9231	5.1343e-02
3	(17)	2.2798e-01	1.5419e-01	0.9231	5.1315e-02
3	(18)	2.2798e-01	1.5419e-01	0.9231	5.1315e-02
3	(19)	2.2798e-01	1.5419e-01	0.9231	5.1315e-02

Caso 3bus_corrections.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 3bus_shunt_fields.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(16)	1.1038e-01	1.8478e-36	1.0522	0.0000e+00
1	(17)	9.7961e-02	1.8478e-36	1.0397	0.0000e+00
1	(18)	9.7961e-02	1.8478e-36	1.0397	0.0000e+00
1	(19)	9.7961e-02	1.8478e-36	1.0397	0.0000e+00
2	(16)	1.0564e-01	4.6161e-02	1.0062	3.8981e-02
2	(17)	1.0788e-01	3.1602e-02	1.0084	2.4423e-02
2	(18)	1.0788e-01	3.1602e-02	1.0084	2.4423e-02
2	(19)	1.0788e-01	3.1602e-02	1.0084	2.4423e-02
3	(16)	1.1425e-01	2.2512e-02	1.0143	-1.9622e-02
3	(17)	1.1982e-01	1.4338e-02	1.0198	-3.7784e-02
3	(18)	1.1982e-01	1.4338e-02	1.0198	-3.7784e-02
3	(19)	1.1982e-01	1.4338e-02	1.0198	-3.7784e-02

Caso 3busfrank.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(17)	8.1110e-02	4.5139e-36	1.0189	0.0000e+00
1	(18)	8.1110e-02	4.5139e-36	1.0189	0.0000e+00

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(19)	8.1109e-02	4.5139e-36	1.0189	0.0000e+00
2	(17)	8.7075e-02	1.8667e-02	1.0129	-1.8667e-02
2	(18)	8.7075e-02	1.8667e-02	1.0129	-1.8667e-02
2	(19)	8.7075e-02	1.8667e-02	1.0129	-1.8667e-02
3	(17)	8.9594e-02	1.2202e-02	0.9527	-2.7882e-02
3	(18)	8.9594e-02	1.2202e-02	0.9527	-2.7882e-02
3	(19)	8.9593e-02	1.2202e-02	0.9527	-2.7881e-02

Caso 3busfrank_continuous_shunt.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(17)	7.5330e-02	1.5742e-35	1.0247	0.0000e+00
1	(18)	7.5330e-02	1.5742e-35	1.0247	0.0000e+00
1	(19)	7.5330e-02	1.5742e-35	1.0247	0.0000e+00
2	(17)	7.9967e-02	7.0006e-03	1.0200	-7.0006e-03
2	(18)	7.9967e-02	7.0006e-03	1.0200	-7.0006e-03
2	(19)	7.9968e-02	7.0006e-03	1.0200	-7.0006e-03
3	(17)	7.9242e-02	4.7808e-03	0.9833	-3.7295e-03
3	(18)	7.9242e-02	4.7808e-03	0.9833	-3.7295e-03
3	(19)	7.9242e-02	4.7808e-03	0.9833	-3.7295e-03

Caso 3busfrank_qlim.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(16)	4.7408e-02	3.9120e-35	1.0526	0.0000e+00
1	(17)	4.6014e-02	3.9120e-35	1.0750	0.0000e+00
1	(18)	4.6014e-02	3.9120e-35	1.0750	0.0000e+00
1	(19)	4.6014e-02	3.9120e-35	1.0750	0.0000e+00
2	(16)	5.5233e-02	3.7207e-02	1.0448	-3.7207e-02
2	(17)	5.6100e-02	3.9834e-02	1.0439	-1.0585e-02
2	(18)	5.6100e-02	3.9834e-02	1.0439	-1.0585e-02
2	(19)	5.6100e-02	3.9834e-02	1.0439	-1.0584e-02
3	(16)	6.0601e-02	1.3612e-02	0.9000	3.0818e-02
3	(17)	4.7995e-02	1.6495e-02	0.9126	4.3346e-02
3	(18)	4.7995e-02	1.6495e-02	0.9126	4.3346e-02
3	(19)	4.7995e-02	1.6495e-02	0.9126	4.3346e-02

Caso 4busfrank_vlim.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 500bus.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(16)	2.8978e-04	1.1110e-03	1.0117	-1.5096e-01
1	(17)	8.2670e-03	8.4292e-03	1.0032	-1.5828e-01
1	(18)	8.2670e-03	8.4292e-03	1.0032	-1.5828e-01
1	(19)	8.2677e-03	8.4295e-03	1.0032	-1.5828e-01
2	(16)	2.9001e-04	1.1103e-03	1.0112	-1.5221e-01
2	(17)	8.2718e-03	8.4500e-03	1.0026	-1.5955e-01

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
2	(18)	8.2718e-03	8.4500e-03	1.0026	-1.5955e-01
2	(19)	8.2725e-03	8.4503e-03	1.0026	-1.5955e-01
3	(16)	6.2408e-05	1.0686e-03	1.0038	-1.7382e-01
3	(17)	3.4492e-03	9.1475e-03	1.0003	-1.8190e-01
3	(18)	3.4492e-03	9.1475e-03	1.0003	-1.8190e-01
3	(19)	3.4495e-03	9.1478e-03	1.0003	-1.8190e-01
4	(16)	6.2967e-05	1.0681e-03	1.0016	-1.7822e-01
4	(17)	3.4568e-03	9.1779e-03	0.9981	-1.8633e-01
4	(18)	3.4568e-03	9.1779e-03	0.9981	-1.8633e-01
4	(19)	3.4570e-03	9.1782e-03	0.9981	-1.8633e-01
5	(16)	2.3079e-04	1.0398e-03	1.0145	1.5011e-02
5	(17)	1.6366e-02	9.0946e-04	0.9983	1.5141e-02
5	(18)	1.6366e-02	9.0946e-04	0.9983	1.5141e-02
5	(19)	1.6366e-02	9.0975e-04	0.9983	1.5141e-02

(mostrando 20 de 2000 barras divergentes; tabela truncada)

Caso 5busfrank.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(17)	6.2160e-02	4.5139e-36	1.0378	0.0000e+00
1	(18)	6.2160e-02	4.5139e-36	1.0378	0.0000e+00
1	(19)	6.2159e-02	4.5139e-36	1.0378	0.0000e+00
2	(17)	6.7734e-02	1.9556e-02	1.0323	-1.9556e-02
2	(18)	6.7734e-02	1.9556e-02	1.0323	-1.9556e-02
2	(19)	6.7733e-02	1.9556e-02	1.0323	-1.9556e-02
3	(17)	6.9520e-02	1.1515e-02	0.9679	-2.4182e-02
3	(18)	6.9520e-02	1.1515e-02	0.9679	-2.4182e-02
3	(19)	6.9519e-02	1.1515e-02	0.9679	-2.4182e-02
4	(17)	7.0209e-02	1.0522e-02	0.9579	-1.6588e-02
4	(18)	7.0209e-02	1.0522e-02	0.9579	-1.6588e-02
4	(19)	7.0209e-02	1.0522e-02	0.9579	-1.6588e-02
5	(17)	7.0553e-02	1.0011e-02	0.9529	-1.2731e-02
5	(18)	7.0553e-02	1.0011e-02	0.9529	-1.2731e-02
5	(19)	7.0553e-02	1.0011e-02	0.9529	-1.2731e-02

Caso 5busfrank_cphs.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
2	(17)	5.9692e-02	5.1113e-02	1.0403	-5.1113e-02
2	(18)	5.9692e-02	5.1113e-02	1.0403	-5.1113e-02
2	(19)	5.9691e-02	5.1114e-02	1.0403	-5.1114e-02
3	(17)	7.0740e-02	2.9323e-02	0.9625	-5.4916e-02
3	(18)	7.0740e-02	2.9323e-02	0.9625	-5.4916e-02
3	(19)	7.0740e-02	2.9323e-02	0.9625	-5.4916e-02
4	(17)	7.2915e-02	2.7677e-02	0.9331	-4.2813e-02
4	(18)	7.2915e-02	2.7677e-02	0.9331	-4.2813e-02
4	(19)	7.2914e-02	2.7677e-02	0.9331	-4.2813e-02
5	(17)	7.4019e-02	2.6779e-02	0.9183	-3.6470e-02
5	(18)	7.4019e-02	2.6779e-02	0.9183	-3.6470e-02
5	(19)	7.4019e-02	2.6779e-02	0.9183	-3.6471e-02

Caso 5busfrank_csca.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 5busfrank_ctaf.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
2	(17)	5.9692e-02	5.1113e-02	1.0403	-5.1113e-02
2	(18)	8.7415e-02	2.1727e-02	1.0126	-2.1727e-02
2	(19)	8.7415e-02	2.1727e-02	1.0126	-2.1727e-02
3	(17)	7.0740e-02	2.9323e-02	0.9625	-5.4916e-02
3	(18)	8.5366e-02	1.5249e-02	0.9478	-4.0841e-02
3	(19)	8.5366e-02	1.5249e-02	0.9478	-4.0842e-02
4	(17)	7.2915e-02	2.7677e-02	0.9331	-4.2813e-02
4	(18)	6.4726e-02	1.4963e-02	0.9922	-3.0099e-02
4	(19)	6.4726e-02	1.4963e-02	0.9922	-3.0099e-02
5	(17)	7.4019e-02	2.6779e-02	0.9183	-3.6470e-02
5	(18)	7.6832e-02	1.4932e-02	0.9894	-2.4624e-02
5	(19)	7.6832e-02	1.4932e-02	0.9894	-2.4624e-02

Caso 5busfrank_ctap.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
2	(17)	5.9692e-02	5.1113e-02	1.0403	-5.1113e-02
2	(18)	8.7415e-02	2.1727e-02	1.0126	-2.1727e-02
2	(19)	8.7415e-02	2.1727e-02	1.0126	-2.1727e-02
3	(17)	7.0740e-02	2.9323e-02	0.9625	-5.4916e-02
3	(18)	8.5366e-02	1.5249e-02	0.9478	-4.0841e-02
3	(19)	8.5366e-02	1.5249e-02	0.9478	-4.0842e-02
4	(17)	7.2915e-02	2.7677e-02	0.9331	-4.2813e-02
4	(18)	6.4726e-02	1.4963e-02	0.9922	-3.0099e-02
4	(19)	6.4726e-02	1.4963e-02	0.9922	-3.0099e-02
5	(17)	7.4019e-02	2.6779e-02	0.9183	-3.6470e-02
5	(18)	7.6832e-02	1.4932e-02	0.9894	-2.4624e-02
5	(19)	7.6832e-02	1.4932e-02	0.9894	-2.4624e-02

Caso 9bus.pwf

Barra	Script	max abs(Δvm)	max abs(Δva)	vm (pu)	va (rad)
1	(17)	9.1552e-02	1.1456e-29	1.0084	0.0000e+00
1	(18)	9.1552e-02	1.1456e-29	1.0084	0.0000e+00
1	(19)	9.1552e-02	1.1456e-29	1.0084	0.0000e+00
2	(17)	9.2661e-02	1.4438e-02	1.0049	-3.6885e-02
2	(18)	9.2661e-02	1.4438e-02	1.0049	-3.6885e-02
2	(19)	9.2661e-02	1.4438e-02	1.0049	-3.6885e-02
3	(17)	8.0620e-02	2.5995e-02	1.0070	-2.9540e-02
3	(18)	8.0620e-02	2.5995e-02	1.0070	-2.9540e-02
3	(19)	8.0620e-02	2.5995e-02	1.0070	-2.9540e-02
4	(17)	9.7705e-02	1.0509e-02	0.9991	-8.1798e-02
4	(18)	9.7705e-02	1.0509e-02	0.9991	-8.1798e-02
4	(19)	9.7705e-02	1.0509e-02	0.9991	-8.1798e-02
5	(17)	1.0304e-01	2.0319e-02	0.9725	-1.5478e-01
5	(18)	1.0304e-01	2.0319e-02	0.9725	-1.5478e-01

Barra	Script	max abs(Δv_m)	max abs(Δv_a)	vm (pu)	va (rad)
5	(19)	1.0304e-01	2.0319e-02	0.9725	-1.5478e-01
6	(17)	9.9268e-02	1.7255e-02	0.9880	-1.3418e-01
6	(18)	9.9268e-02	1.7255e-02	0.9880	-1.3418e-01
6	(19)	9.9268e-02	1.7255e-02	0.9880	-1.3418e-01
7	(17)	9.6067e-02	1.2155e-02	1.0039	-9.2674e-02
7	(18)	9.6067e-02	1.2155e-02	1.0039	-9.2674e-02

(mostrando 20 de 27 barras divergentes; tabela truncada)

Caso 9bus_transformer_fields.pwf

Barra	Script	max abs(Δv_m)	max abs(Δv_a)	vm (pu)	va (rad)
1	(17)	6.4801e-02	1.0160e-30	1.0352	0.0000e+00
1	(18)	8.9368e-02	1.0160e-30	1.0106	0.0000e+00
1	(19)	8.9368e-02	1.0160e-30	1.0106	0.0000e+00
2	(17)	8.3621e-02	1.9529e-02	1.0164	-3.1430e-02
2	(18)	9.6048e-02	2.5310e-02	1.0039	-2.5650e-02
2	(19)	9.6048e-02	2.5310e-02	1.0039	-2.5650e-02
3	(17)	7.7793e-02	3.0519e-02	1.0204	-2.4604e-02
3	(18)	8.0493e-02	3.7742e-02	1.0177	-1.7382e-02
3	(19)	8.0492e-02	3.7742e-02	1.0177	-1.7382e-02
4	(17)	7.5034e-02	6.3924e-03	1.0237	-7.7681e-02
4	(18)	5.5490e-02	3.1396e-03	1.0433	-7.4428e-02
4	(19)	5.5490e-02	3.1396e-03	1.0433	-7.4428e-02
5	(17)	8.3040e-02	1.2468e-02	0.9953	-1.4693e-01
5	(18)	6.6648e-02	6.3032e-03	1.0117	-1.4077e-01
5	(19)	6.6648e-02	6.3032e-03	1.0117	-1.4077e-01
6	(17)	8.0878e-02	1.0274e-02	1.0107	-1.2720e-01
6	(18)	6.6504e-02	4.5117e-03	1.0251	-1.2144e-01
6	(19)	6.6503e-02	4.5117e-03	1.0251	-1.2144e-01
7	(17)	8.4597e-02	1.0487e-02	1.0192	-8.5757e-02
7	(18)	7.6845e-02	1.6990e-02	1.0270	-7.9254e-02

(mostrando 20 de 27 barras divergentes; tabela truncada)

Caso test_defaults.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso test_line_shunt.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso test_system.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

3. Fluxos nos Ramos

Mesmo critério da Seção 2 aplicado a pf_pu, qf_pu, pt_pu, qt_pu (excede 0.0001 em AMBAS (14) E (15)).

Caso 01 MAXIMA NOTURNA_DEZ25.PWF

Sem referência convergida; análise pulada.

Caso 300bus.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(16)	8.7637e-03	1.6722e-02	-0.0967	-0.1066
1	(17)	2.6654e-03	1.1404e-02	-0.0906	-0.1012
1	(18)	6.4154e-02	5.8144e-02	-0.1521	-0.1480
1	(19)	6.4141e-02	5.9701e-02	-0.1521	-0.1495
2	(16)	4.2944e-04	2.5047e-02	0.0096	-0.0216
2	(17)	2.0825e-04	1.9776e-02	0.0093	-0.0163
2	(18)	1.9233e-05	8.4266e-04	0.0091	0.0026
2	(19)	1.8772e-05	7.8702e-04	0.0091	0.0027
3	(16)	7.3664e-03	8.2470e-02	4.0700	1.1219
3	(17)	4.4326e-03	1.3704e-02	4.0671	1.0531
3	(18)	1.9449e-02	2.8987e-01	4.0821	1.3293
3	(19)	1.9335e-02	3.0189e-01	4.0820	1.3413
4	(16)	1.9169e-03	2.0617e-01	-0.3065	-0.1929
4	(17)	2.9446e-03	2.2058e-01	-0.3055	-0.2073
4	(18)	5.2350e-02	3.5533e-01	-0.2561	-0.3421
4	(19)	4.9301e-02	3.5728e-01	-0.2591	-0.3440
5	(16)	2.4125e-04	2.0302e-02	0.2835	-0.2344
5	(17)	2.4565e-04	2.4406e-02	0.2835	-0.2385
5	(18)	9.3041e-05	1.6334e-03	0.2836	-0.2157
5	(19)	6.0376e-05	1.0702e-03	0.2836	-0.2151

(mostrando 20 de 1642 ramos divergentes; tabela truncada)

Caso 3bus.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	1.7942e-02	2.0243e-02	0.0179	-0.0202
1	(18)	1.7942e-02	2.0243e-02	0.0179	-0.0202
1	(19)	1.7942e-02	2.0243e-02	0.0179	-0.0202
2	(17)	9.6216e-03	8.8557e-03	0.1653	0.0214
2	(18)	9.6216e-03	8.8557e-03	0.1653	0.0214
2	(19)	9.6216e-03	8.8558e-03	0.1653	0.0214
3	(17)	8.0022e-03	1.0413e-02	0.1477	0.0407
3	(18)	8.0022e-03	1.0413e-02	0.1477	0.0407
3	(19)	8.0022e-03	1.0413e-02	0.1477	0.0407

Caso 3bus_DBSH.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	2.5528e-02	2.6355e-02	0.0255	-0.0264
1	(18)	2.5528e-02	2.6355e-02	0.0255	-0.0264
1	(19)	2.5527e-02	2.6337e-02	0.0255	-0.0263
2	(17)	1.2565e-02	4.1940e-02	0.1786	0.2555
2	(18)	1.2565e-02	4.1940e-02	0.1786	0.2555
2	(19)	1.2565e-02	4.1948e-02	0.1786	0.2555
3	(17)	1.0820e-02	6.7120e-02	0.1552	0.2807
3	(18)	1.0820e-02	6.7120e-02	0.1552	0.2807
3	(19)	1.0821e-02	6.7111e-02	0.1552	0.2806

Caso 3bus_DCER.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	2.7124e-02	7.3833e-02	0.0271	-0.0738
1	(18)	2.7124e-02	7.3833e-02	0.0271	-0.0738
1	(19)	2.7124e-02	7.3834e-02	0.0271	-0.0738
2	(17)	1.1152e-02	2.2511e-02	0.1772	0.1999
2	(18)	1.1152e-02	2.2511e-02	0.1772	0.1999
2	(19)	1.1152e-02	2.2512e-02	0.1772	0.1999
3	(17)	1.0094e-02	5.9687e-02	0.1560	0.2732
3	(18)	1.0094e-02	5.9687e-02	0.1560	0.2732
3	(19)	1.0094e-02	5.9687e-02	0.1560	0.2732

Caso 3bus_DCSC.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(16)	8.6927e-02	1.0842e-01	0.0872	0.0131
1	(17)	8.8753e-02	1.1063e-01	0.0890	0.0153
1	(18)	8.8753e-02	1.1063e-01	0.0890	0.0153
1	(19)	8.8753e-02	1.1063e-01	0.0890	0.0153
2	(16)	8.9280e-02	1.0857e-01	0.0895	0.0167
2	(17)	8.7626e-02	1.0757e-01	0.0879	0.0157
2	(18)	8.7626e-02	1.0757e-01	0.0879	0.0157
2	(19)	8.7626e-02	1.0757e-01	0.0879	0.0157
3	(16)	4.0368e-02	1.6251e-01	-0.0404	0.1625
3	(17)	4.2022e-02	1.8236e-01	-0.0420	-0.0208
3	(18)	4.2022e-02	1.8236e-01	-0.0420	-0.0208
3	(19)	4.2022e-02	1.8236e-01	-0.0420	-0.0208

Caso 3bus_DCline.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(16)	1.6337e-02	1.1549e+00	0.1721	1.1852
1	(17)	1.6337e-02	1.1549e+00	0.1721	1.1852
1	(18)	1.6337e-02	1.1549e+00	0.1721	1.1852
1	(19)	1.6337e-02	1.1549e+00	0.1721	1.1852
2	(16)	6.1106e+00	5.2529e-01	6.2663	0.5556
2	(17)	6.1106e+00	5.2529e-01	6.2663	0.5556
2	(18)	6.1106e+00	5.2529e-01	6.2663	0.5556
2	(19)	6.1106e+00	5.2529e-01	6.2663	0.5556

Caso 3bus_DSHL.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(16)	4.6185e-01	9.7792e-01	0.1160	0.8040
1	(17)	4.6181e-01	9.7779e-01	0.1160	0.8041
1	(18)	4.6181e-01	9.7779e-01	0.1160	0.8041
1	(19)	4.6181e-01	9.7779e-01	0.1160	0.8041
2	(16)	3.9122e-01	9.0884e-01	0.2354	0.5000
2	(17)	3.9122e-01	9.0897e-01	0.2354	0.4999
2	(18)	3.9122e-01	9.0897e-01	0.2354	0.4999

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
2	(19)	3.9122e-01	9.0897e-01	0.2354	0.4999
3	(16)	2.8798e-02	1.1972e+00	0.1367	2.4275
3	(17)	2.8800e-02	1.1970e+00	0.1367	2.4272
3	(18)	2.8800e-02	1.1970e+00	0.1367	2.4272
3	(19)	2.8800e-02	1.1970e+00	0.1367	2.4272

Caso 3bus_corrections.pwf

Sem referência convergida; análise pulada.

Caso 3bus_shunt_fields.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(16)	1.1099e-02	2.7888e-01	0.0274	0.2543
1	(17)	7.4807e-03	1.8892e-01	0.0238	0.1643
1	(18)	7.4807e-03	1.8892e-01	0.0238	0.1643
1	(19)	7.4807e-03	1.8892e-01	0.0238	0.1643
2	(16)	6.5786e-03	1.3363e-01	0.1729	0.0505
2	(17)	5.8489e-03	1.0796e-01	0.1721	-0.0558
2	(18)	5.8489e-03	1.0796e-01	0.1721	-0.0558
2	(19)	5.8489e-03	1.0796e-01	0.1721	-0.0558
3	(16)	5.0589e-03	1.2965e-01	0.1466	-0.1893
3	(17)	2.5992e-03	1.5028e-01	0.1491	-0.2099
3	(18)	2.5992e-03	1.5028e-01	0.1491	-0.2099
3	(19)	2.5992e-03	1.5028e-01	0.1491	-0.2099

Caso 3busfrank.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	7.1144e-02	4.1374e-02	0.0711	-0.0380
1	(18)	7.1144e-02	4.1374e-02	0.0711	-0.0380
1	(19)	7.1144e-02	4.1375e-02	0.0711	-0.0380
2	(17)	3.7128e-02	2.0680e-02	0.2695	0.1096
2	(18)	3.7128e-02	2.0680e-02	0.2695	0.1096
2	(19)	3.7128e-02	2.0680e-02	0.2695	0.1096
3	(17)	3.2472e-02	2.0374e-02	0.1999	0.1431
3	(18)	3.2472e-02	2.0374e-02	0.1999	0.1431
3	(19)	3.2472e-02	2.0375e-02	0.1999	0.1431

Caso 3busfrank_continuous_shunt.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	3.3914e-02	3.3591e-02	0.0339	-0.0078
1	(18)	3.3914e-02	3.3591e-02	0.0339	-0.0078
1	(19)	3.3914e-02	3.3591e-02	0.0339	-0.0078
2	(17)	3.3914e-02	3.3591e-02	0.0339	-0.0078
2	(18)	3.3914e-02	3.3591e-02	0.0339	-0.0078
2	(19)	3.3914e-02	3.3591e-02	0.0339	-0.0078
3	(17)	1.7181e-02	1.6081e-02	0.1322	0.1063
3	(18)	1.7181e-02	1.6081e-02	0.1322	0.1063

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
3	(19)	1.7181e-02	1.6081e-02	0.1322	0.1063
4	(17)	1.7181e-02	1.6081e-02	0.1322	0.1063
4	(18)	1.7181e-02	1.6081e-02	0.1322	0.1063
4	(19)	1.7181e-02	1.6081e-02	0.1322	0.1063
5	(17)	1.6385e-02	1.7589e-02	0.0987	0.1130
5	(18)	1.6385e-02	1.7589e-02	0.0987	0.1130
5	(19)	1.6385e-02	1.7590e-02	0.0987	0.1130
6	(17)	1.6385e-02	1.7589e-02	0.0987	0.1130
6	(18)	1.6385e-02	1.7589e-02	0.0987	0.1130
6	(19)	1.6385e-02	1.7590e-02	0.0987	0.1130

Caso 3busfrank_qlim.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(16)	1.3875e-01	9.2623e-02	0.1388	-0.0926
1	(17)	1.2901e-01	2.1089e-01	0.1290	0.0584
1	(18)	1.2901e-01	2.1089e-01	0.1290	0.0584
1	(19)	1.2901e-01	2.1089e-01	0.1290	0.0584
2	(16)	6.8166e-02	3.2490e-02	0.3819	0.5284
2	(17)	7.2877e-02	1.0193e-01	0.3866	0.6065
2	(18)	7.2877e-02	1.0193e-01	0.3866	0.6065
2	(19)	7.2877e-02	1.0193e-01	0.3866	0.6065
3	(16)	4.9630e-02	4.6112e-02	0.2641	0.6070
3	(17)	5.7967e-02	1.1532e-01	0.2558	0.5296
3	(18)	5.7967e-02	1.1532e-01	0.2558	0.5296
3	(19)	5.7967e-02	1.1532e-01	0.2558	0.5296

Caso 4busfrank_vlim.pwf

Sem referência convergida; análise pulada.

Caso 500bus.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(16)	9.2514e-05	3.4276e-05	-0.8760	-0.0131
1	(17)	6.8737e-03	4.8366e-02	-0.8828	-0.0615
1	(18)	6.8737e-03	4.8366e-02	-0.8828	-0.0615
1	(19)	6.8737e-03	4.8365e-02	-0.8828	-0.0615
2	(16)	1.4068e-07	2.4088e-04	-1.0350	-0.2761
2	(17)	1.4068e-07	1.3682e-05	-1.0350	-0.2759
2	(18)	1.4068e-07	1.3682e-05	-1.0350	-0.2759
2	(19)	1.5068e-07	1.3684e-05	-1.0350	-0.2759
5	(16)	1.1114e-04	5.5491e-04	-1.8988	-0.1657
5	(17)	2.7802e-03	5.2124e-02	-1.9015	-0.2173
5	(18)	2.7802e-03	5.2124e-02	-1.9015	-0.2173
5	(19)	2.7802e-03	5.2126e-02	-1.9015	-0.2173
7	(16)	5.3678e-08	1.7636e-04	-0.2561	-0.0685
8	(16)	9.1236e-04	5.1421e-03	0.4230	0.3333
8	(17)	1.1821e-02	7.0981e-02	0.4102	0.2572
8	(18)	1.1821e-02	7.0981e-02	0.4102	0.2572
8	(19)	1.1822e-02	7.0982e-02	0.4102	0.2572
10	(16)	4.4279e-05	2.2927e-04	0.3893	0.1361

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
10	(17)	1.3591e-03	4.3176e-03	0.3880	0.1320
10	(18)	1.3591e-03	4.3176e-03	0.3880	0.1320

(mostrando 20 de 1894 ramos divergentes; tabela truncada)

Caso 5busfrank.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
2	(17)	7.5027e-02	4.3620e-02	0.0750	-0.0436
2	(18)	7.5027e-02	4.3620e-02	0.0750	-0.0436
2	(19)	7.5027e-02	4.3620e-02	0.0750	-0.0436
4	(17)	3.8481e-02	1.9498e-02	0.2764	0.1315
4	(18)	3.8481e-02	1.9498e-02	0.2764	0.1315
4	(19)	3.8481e-02	1.9498e-02	0.2764	0.1315
5	(17)	3.4259e-02	2.1394e-02	0.2037	0.1702
5	(18)	3.4259e-02	2.1394e-02	0.2037	0.1702
5	(19)	3.4259e-02	2.1394e-02	0.2037	0.1702

Caso 5busfrank_cphs.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
2	(17)	1.2157e-01	1.8530e-01	0.1216	-0.1853
2	(18)	1.2157e-01	1.8530e-01	0.1216	-0.1853
2	(19)	1.2157e-01	1.8530e-01	0.1216	-0.1853
4	(17)	5.8603e-02	8.3310e-02	0.3509	0.0370
4	(18)	5.8603e-02	8.3310e-02	0.3509	0.0370
4	(19)	5.8603e-02	8.3312e-02	0.3509	0.0370
5	(17)	4.9264e-02	9.2288e-02	0.2431	0.2126
5	(18)	4.9264e-02	9.2288e-02	0.2431	0.2126
5	(19)	4.9264e-02	9.2290e-02	0.2431	0.2126

Caso 5busfrank_csca.pwf

Sem referência convergida; análise pulada.

Caso 5busfrank_ctaf.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	1.2157e-01	1.8530e-01	0.1216	-0.1853
1	(18)	1.1169e-01	1.0481e-01	0.1117	-0.0177
1	(19)	1.1169e-01	1.0481e-01	0.1117	-0.0177
2	(17)	5.8603e-02	8.3310e-02	0.3509	0.0370
2	(18)	5.9738e-02	5.0805e-02	0.3521	0.1184
2	(19)	5.9738e-02	5.0804e-02	0.3521	0.1184
3	(17)	4.9264e-02	9.2288e-02	0.2431	0.2126
3	(18)	5.2933e-02	5.1987e-02	0.2394	0.1288
3	(19)	5.2933e-02	5.1986e-02	0.2394	0.1288

Caso 5busfrank_ctap.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	1.2157e-01	1.8530e-01	0.1216	-0.1853
1	(18)	1.1169e-01	1.0481e-01	0.1117	-0.0177
1	(19)	1.1169e-01	1.0481e-01	0.1117	-0.0177
2	(17)	5.8603e-02	8.3310e-02	0.3509	0.0370
2	(18)	5.9738e-02	5.0805e-02	0.3521	0.1184
2	(19)	5.9738e-02	5.0804e-02	0.3521	0.1184
3	(17)	4.9264e-02	9.2288e-02	0.2431	0.2126
3	(18)	5.2933e-02	5.1987e-02	0.2394	0.1288
3	(19)	5.2933e-02	5.1986e-02	0.2394	0.1288

Caso 9bus.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	4.4032e-02	2.4064e-02	0.4982	0.0162
1	(18)	4.4032e-02	2.4064e-02	0.4982	0.0162
1	(19)	4.4032e-02	2.4063e-02	0.4982	0.0162
2	(17)	6.0116e-02	6.4742e-02	0.8619	0.1532
2	(18)	6.0116e-02	6.4742e-02	0.8619	0.1532
2	(19)	6.0116e-02	6.4742e-02	0.8619	0.1532
3	(17)	8.7234e-02	4.3266e-02	0.5673	-0.0490
3	(18)	8.7234e-02	4.3266e-02	0.5673	-0.0490
3	(19)	8.7234e-02	4.3266e-02	0.5673	-0.0490
4	(17)	8.6675e-02	1.4857e-02	-0.3382	-0.2228
4	(18)	8.6675e-02	1.4857e-02	-0.3382	-0.2228
4	(19)	8.6675e-02	1.4857e-02	-0.3382	-0.2228
5	(17)	6.3068e-02	5.2152e-02	0.4018	-0.0266
5	(18)	6.3068e-02	5.2152e-02	0.4018	-0.0266
5	(19)	6.3068e-02	5.2151e-02	0.4018	-0.0266
6	(17)	4.4052e-02	2.8729e-02	-0.5039	-0.2034
6	(18)	4.4052e-02	2.8729e-02	-0.5039	-0.2034
6	(19)	4.4052e-02	2.8729e-02	-0.5039	-0.2034
7	(17)	1.4735e-01	1.1386e-01	1.4292	0.2226
7	(18)	1.4735e-01	1.1386e-01	1.4292	0.2226

(mostrando 20 de 27 ramos divergentes; tabela truncada)

Caso 9bus_transformer_fields.pwf

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
1	(17)	4.0424e-02	1.9503e-02	0.4982	0.0047
1	(18)	4.1254e-02	4.5315e-02	0.4974	0.0305
1	(19)	4.1253e-02	4.5313e-02	0.4974	0.0305
2	(17)	5.8693e-02	9.9064e-02	0.8612	0.1777
2	(18)	5.9835e-02	1.3946e-01	0.8600	0.2181
2	(19)	5.9835e-02	1.3946e-01	0.8600	0.2181
3	(17)	8.1980e-02	8.4402e-02	0.5666	-0.0292
3	(18)	8.1283e-02	1.4152e-01	0.5673	0.0280
3	(19)	8.1283e-02	1.4152e-01	0.5673	0.0280
4	(17)	8.1306e-02	6.2034e-02	-0.3387	-0.1941

Ramo	Script	max abs(Δ pf)	max abs(Δ qf)	pf (pu)	qf (pu)
4	(18)	8.0584e-02	1.2479e-01	-0.3379	-0.1313
4	(19)	8.0584e-02	1.2479e-01	-0.3379	-0.1313
5	(17)	6.0969e-02	1.3495e-02	0.4018	-0.0755
5	(18)	6.1798e-02	6.9621e-02	0.4026	-0.1316
5	(19)	6.1798e-02	6.9621e-02	0.4026	-0.1316
6	(17)	4.0446e-02	7.6770e-03	-0.5039	-0.2096
6	(18)	4.1279e-02	2.0400e-02	-0.5047	-0.1817
6	(19)	4.1279e-02	2.0398e-02	-0.5047	-0.1817
7	(17)	1.4067e-01	1.7929e-01	1.4278	0.2618
7	(18)	1.4112e-01	2.7457e-01	1.4273	0.3571

(mostrando 20 de 27 ramos divergentes; tabela truncada)

Caso test_defaults.pwf

Sem referência convergida; análise pulada.

Caso test_line_shunt.pwf

Sem referência convergida; análise pulada.

Caso test_system.pwf

Sem referência convergida; análise pulada.

4. Avaliação de Impacto Sistêmico

Tabela de perdas totais (em pu, na base do caso). $\Delta P_{loss\%}$ calculado em relação a (14) e (15) quando ambas convergiram. Para converter para MW, multiplique pela baseMVA do caso (tipicamente 100 MVA).

Caso	(14)	(15)	(16)	(17)	(18)	(19)	$\Delta\%$ vs(14): 17/18/19	$\Delta\%$ vs(15): 17/18/19
01 MAXIMA NOTURNA_DEZ25.PWF	—	—	—	—	—	—	— / — / —	— / — / —
300bus.pwf	—	5.77785	5.66572	5.67328	5.93561	5.93797	— / — / —	-1.81% / +2.73% / +2.77%
3bus.pwf	0.00753	0.00890	0.00890	0.00928	0.00928	0.00928	+23.25% / +23.25% / +23.25%	+4.24% / +4.24% / +4.24%
3bus_DBSH.pwf	0.02819	0.02919	0.02919	0.03014	0.03014	0.03014	+6.90% / +6.90% / +6.90%	+3.27% / +3.27% / +3.27%
3bus_DCER.pwf	0.02819	0.02919	0.02919	0.03032	0.03032	0.03032	+7.55% / +7.55% / +7.55%	+3.90% / +3.90% / +3.90%
3bus_DCSC.pwf	0.00006	0.00575	0.00271	0.00288	0.00288	0.00288	+4692.92% / +4692.92% / +4692.92%	-49.87% / -49.87% / -49.87%

Caso	(14)	(15)	(16)	(17)	(18)	(19)	$\Delta\%$ vs(14): 17/18/19	$\Delta\%$ vs(15): 17/18/19
3bus_DCline.pwf	0.00753	—	6.13444	6.13444	6.13444	6.13444	+81387.72% / — +81387.72% /	— / — / —
3bus_DSHL.pwf	1.03045	0.17739	0.17742	0.17742	0.17742	0.17742	+81387.72% — / — / —	-82.78% / -82.78% / -82.78%
3bus_corrections.pwf	—	—	—	—	—	—	— / — / —	— / — / —
3bus_shunt_0.02529.pwf	0.01040	0.02624	0.02190	0.02190	0.02190	0.02190	-13.42% / -13.42% / -13.42%	+110.44% / +110.44% /
3busfrank.pwf	0.02076	0.02708	0.02708	0.02655	0.02655	0.02655	+27.89% / +27.89% / +27.89%	-1.94% / -1.94% / -1.94%
3busfrank_continuous_shunt.pwf	0.01616	0.01889	0.01889	0.01817	0.01817	0.01817	+12.42% / +12.42% / +12.42%	-3.82% / -3.82% / -3.82%
3busfrank_qlim.pwf	0.16015	0.14665	0.14162	0.14162	0.14162	0.14162	+14.61% / +14.61% / +14.61%	-11.57% / -11.57% / -11.57%
4busfrank_vlim.pwf	—	0.14588	0.14574	0.14574	0.14574	0.17144	— / — / —	— / — / —
500bus.pwf	—	0.86293	0.86366	0.90431	0.90431	0.90432	— / — / —	+4.80% / +4.80% / +4.80%
5busfrank.pwf	0.02404	0.03130	0.03130	0.02953	0.02953	0.02953	+22.83% / +22.83% / +22.83%	-5.65% / -5.65% / -5.65%
5busfrank_caps.pwf	0.04591	0.04591	0.05071	0.05071	0.05071	0.05071	+53.78% / +53.78% / +53.78%	+10.46% / +10.46% / +10.46%
5busfrank_csca.pwf	—	0.22963	0.17393	0.17393	0.17393	0.17393	— / — / —	— / — / —
5busfrank_cua.pwf	0.04591	0.04591	0.05071	0.04196	0.04196	0.04197	+53.78% / +27.27% / +27.27%	+10.46% / -8.58% / -8.58%
5busfrank_cua2.pwf	0.04591	0.04591	0.05071	0.04196	0.04196	0.04197	+53.78% / +27.27% / +27.27%	+10.46% / -8.58% / -8.58%

Caso	(14)	(15)	(16)	(17)	(18)	(19)	$\Delta\%$ vs(14): 17/18/19	$\Delta\%$ vs(15): 17/18/19
9bus.pwf	0.02316	0.02491	0.02491	0.02917	0.02917	0.02917	+25.96% / +25.96% / +25.96% +20.25% / +18.33% / +18.32%	+17.08% / +17.08% / +17.08% +11.46% / +9.68% / +9.68%
9bus_transformerfield.pwf	0.02316	0.02491	0.02491	0.02777	0.02732	0.02732		
test_defaults.pwf	—	—	—	—	—	—	— / — / —	— / — / —
test_line_shunt.pwf	—	—	—	—	—	—	— / — / —	— / — / —
test_system.pwf	—	—	0.00229	0.00243	0.00243	0.00243	— / — / —	— / — / —

5. Conclusão Técnica

Robustez de convergência (sobre 25 casos):

- (14): 15/25 (60.0%)
- (15): 17/25 (68.0%)
- (16): 21/25 (84.0%)
- (17): 21/25 (84.0%)
- (18): 21/25 (84.0%)
- (19): 21/25 (84.0%)

Interpretação física dos controles:

- **QLIM** (limites de Q dos geradores; introduzido em (16)): impõe o setpoint de tensão das barras PV via slack quadrático `sl_v[i]`. Quando o reativo de máquina satura, a barra migra efetivamente para PQ — visualizável em casos onde `vm_pu` em (16)/(17)/(18) diverge do PF tradicional (15) porque a referência de tensão é abandonada pelo gerador saturado. Esperado: pequena perturbação no perfil de tensão em barras vizinhas a geradores com folga reduzida de Q.
- **CSCA** (controle de susceptância shunt; introduzido em (17)): libera `bs_var[i]` como variável de decisão dentro de `[bsmin, bsmax]` lido do bloco DOPC do PWF. Onde o ANAREDE permite chaveamento (banco capacitivo/reativo), o OPF pode redespachar reativo localmente — tende a aliviar tensão em barras de carga e reduzir geração reativa de máquinas, podendo diminuir perdas. Esperado: ΔP_{loss} negativo (perdas menores) em casos com shunts controláveis significativos.
- **CTAP** (controle de tap em transformadores; introduzido em (18)): libera `tm_var[1]` como contínuo em `[tapmin, tapmax]`. Modifica a relação de transformação para redistribuir fluxos e tensões secundárias. Diferente do CSCA (que adiciona Q local), o CTAP redistribui fluxo ao longo do ramo. Pode competir com o CSCA — ambos atuam sobre tensão, mas com mecanismos físicos distintos. Esperado: efeito de segunda ordem no perfil de tensão e nas perdas; impacto maior em redes com folga de tap.
- **DERA** (corte de carga preventivo; introduzido em (19)): adiciona `corte_p[1]` `[0, pd]` e `corte_q[1]` (vinculado por fator de potência da carga). Penalidade hierárquica por nível de tensão (1e7 para <69 kV, 5e7 para 69–230 kV, 1e8 para >230 kV) faz com que o solver prefira cortar carga na distribuição antes de cortar em EAT. Esperado: corte zero quando a rede está OK; corte positivo apenas em casos onde os outros controles não bastam para satisfazer as restrições.

Interações:

- *CTAP* $\times$ *CSCA*: podem atuar sobre o mesmo objetivo de tensão, mas o solver tende a preferir a manobra de menor “custo” (`PENALIDADE_MENOR` = 1e4 para shunt, sem penalidade para tap) — ou seja, CTAP age primeiro quando disponível.

- *DERA × resto*: DERA tem peso $1e7$ (» PENALIDADE = $1e6$ dos slacks). Logo, o solver só cortará carga quando o resto realmente não conseguir respeitar QLIM/VLIM dentro das margens disponíveis. Casos onde DERA mostra corte > 0 (visíveis pela diferença de carga entre (19) e (14)/(15) na tabela de perdas) indicam estresse efetivo da rede.

Convergência: quanto mais variáveis e slacks o modelo adiciona, mais sensível à inicialização e às tolerâncias o Ipopt fica. Casos pequenos (3bus/5bus) convergem facilmente em todas as formulações; redes maiores (300bus, 500bus, SIN) podem expor diferenças de robustez entre (14)/(15) (PowerModels) e as variantes JuMP (16)–(19). Iter count não foi capturado nesta bateria (Ipopt MOI não expôs **BarrierIterations** para os solves rodados); para análise futura, pode-se parsear o log do Ipopt onde aparece “Number of Iterations....”.